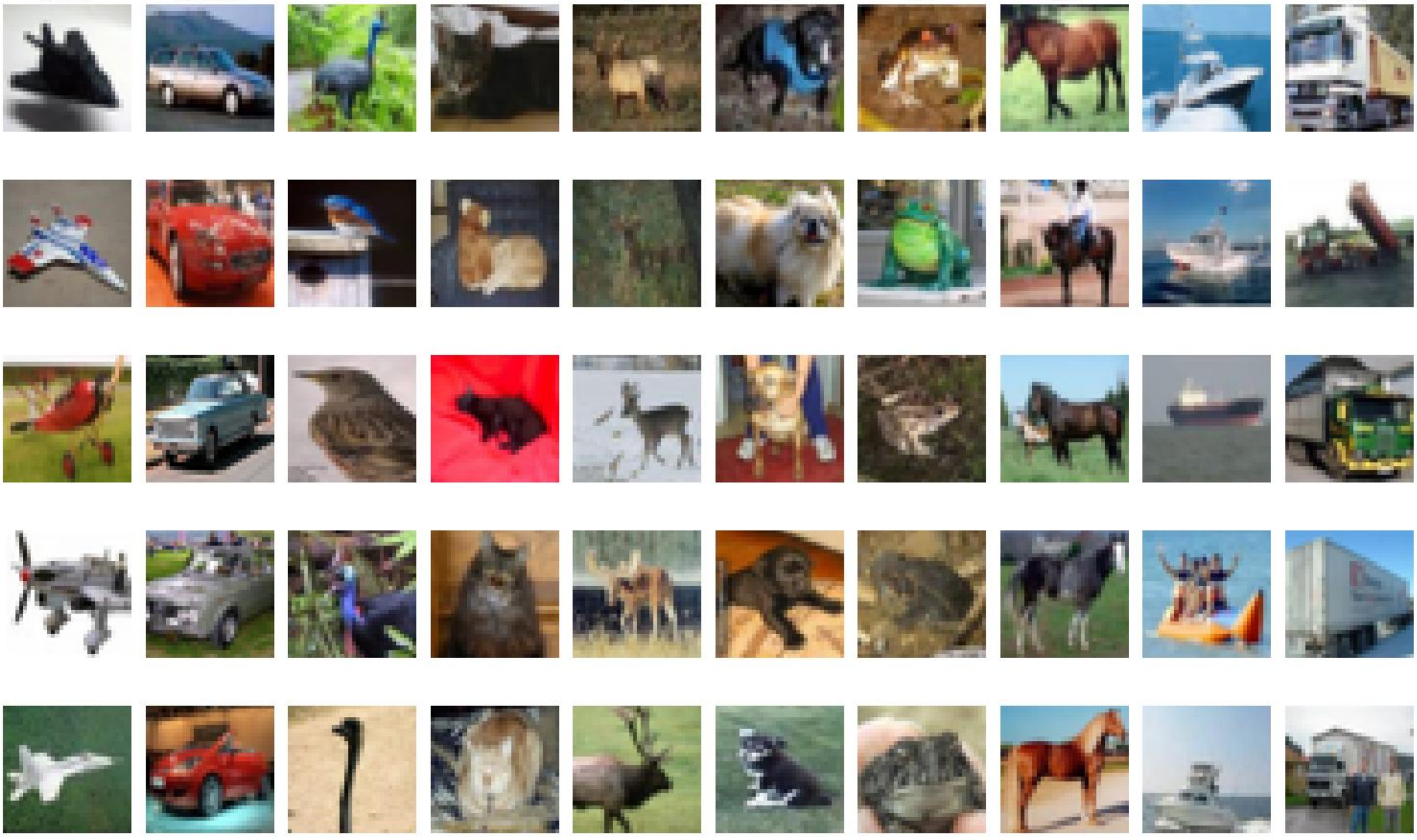
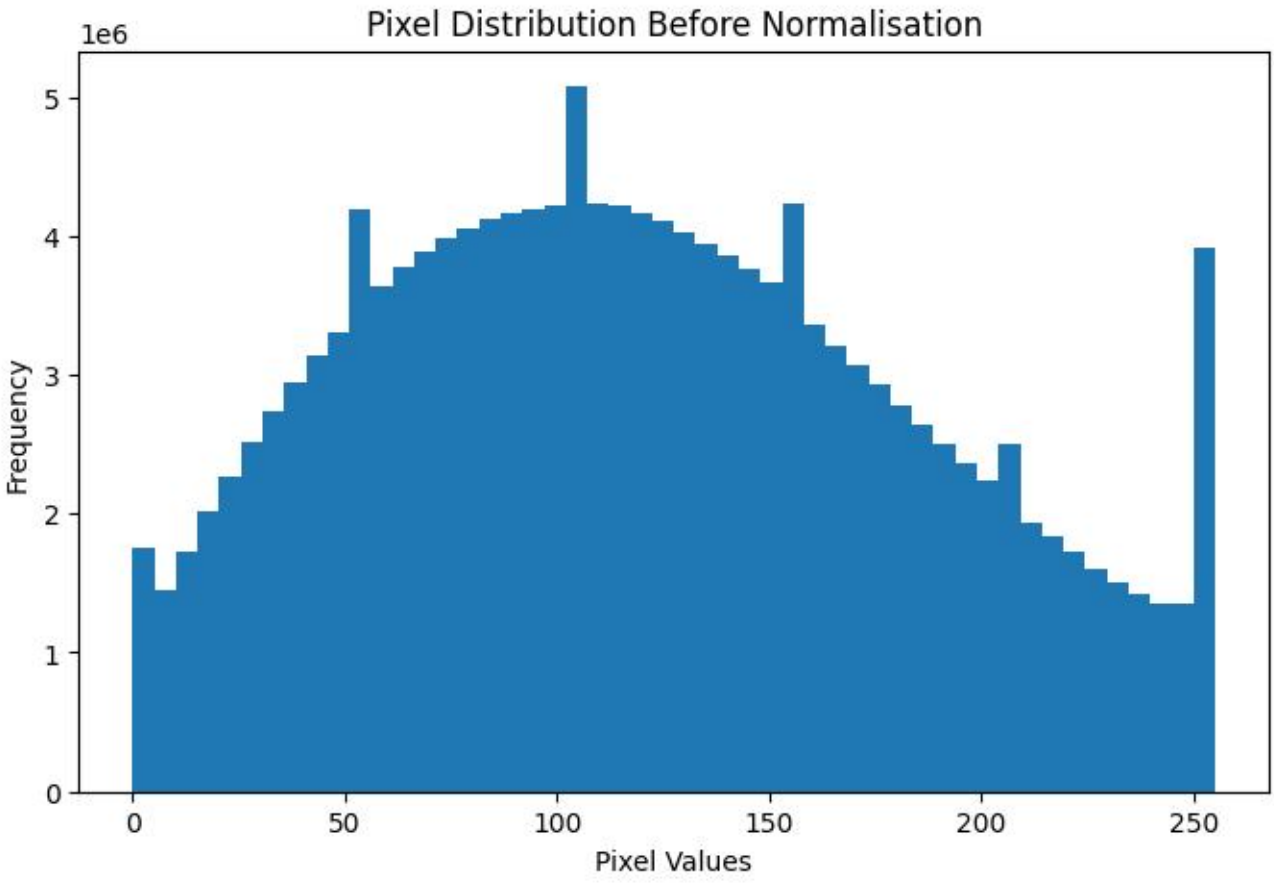
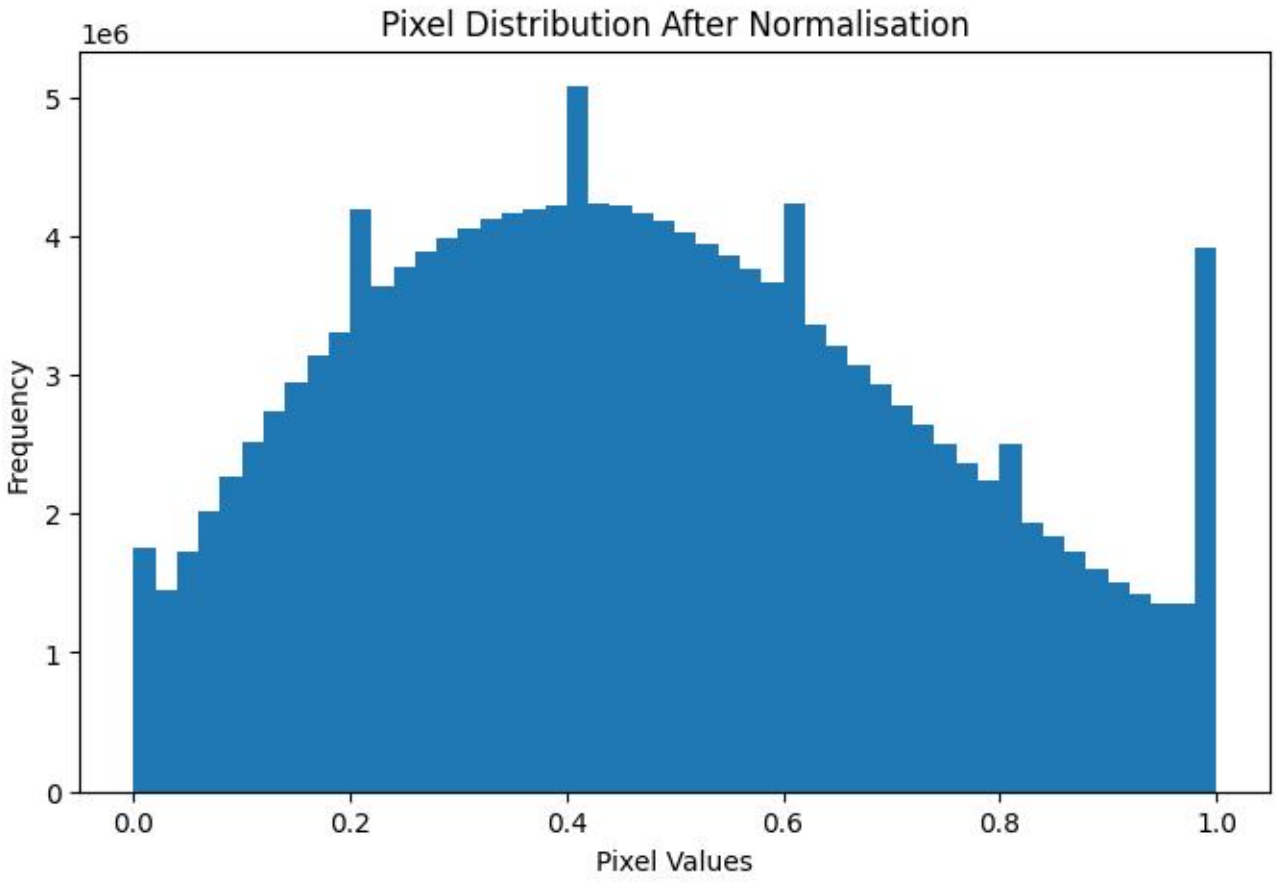
CIFAR-10 – 5 Samples per Class

Airplane | Automobile | Bird | Cat | Deer | Dog | Frog | Horse | Ship | Truck

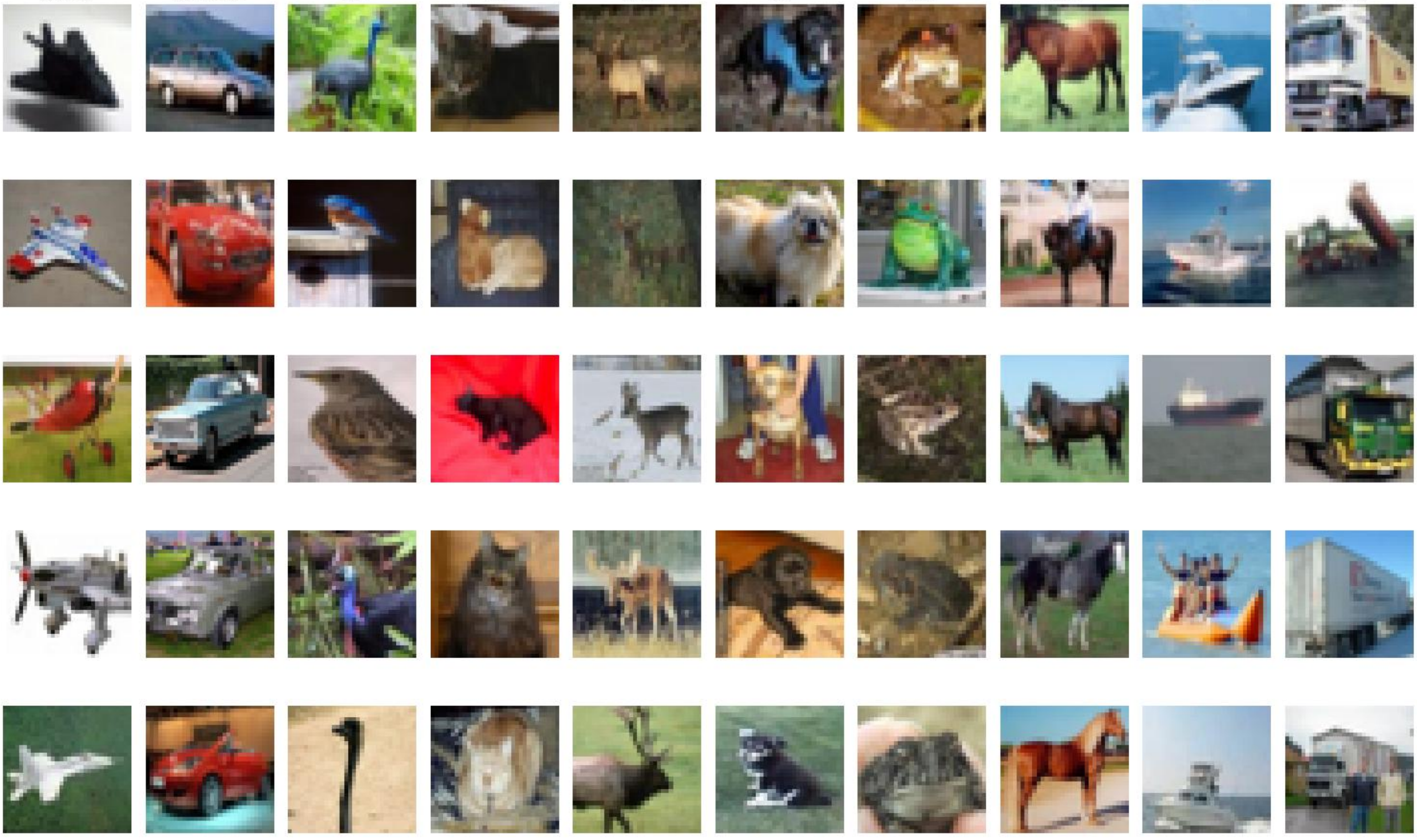



Pixel Distribution Before Normalisation
1e6
5
4
3
2
1
0
Frequency
0
50
100
150
200
250
Pixel Values

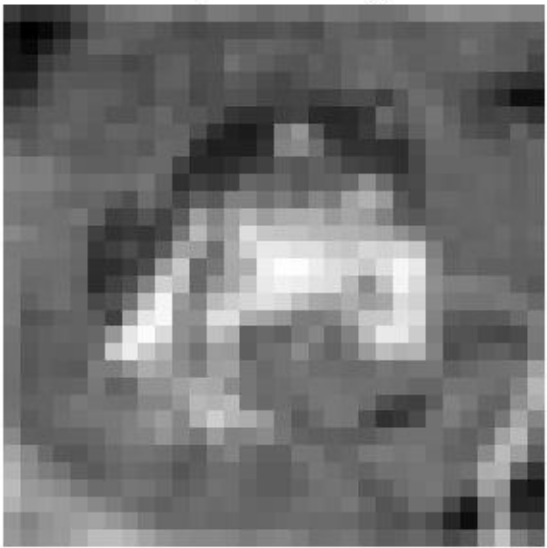

Pixel Distribution After Normalisation
1e6
5
4
3
2
1
0
Frequency
0.0
0.2
0.4
0.6
0.8
1.0
Pixel Values

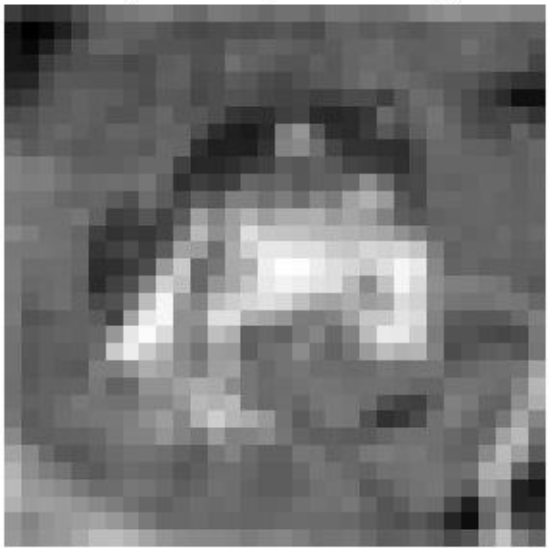
CIFAR-10 – 5 Samples per Class

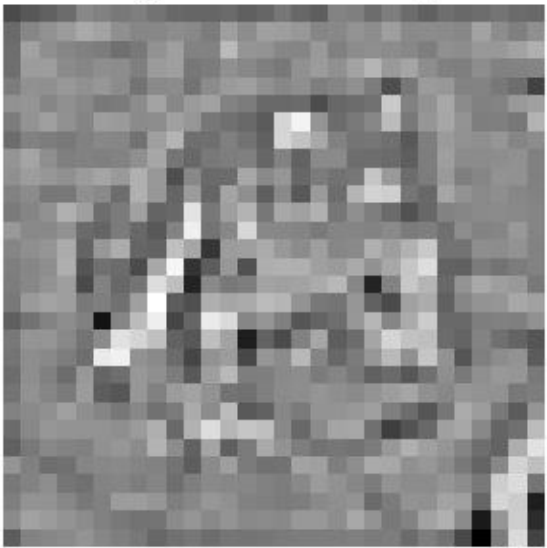
Airplane | Automobile | Bird | Cat | Deer | Dog | Frog | Horse | Ship | Truck

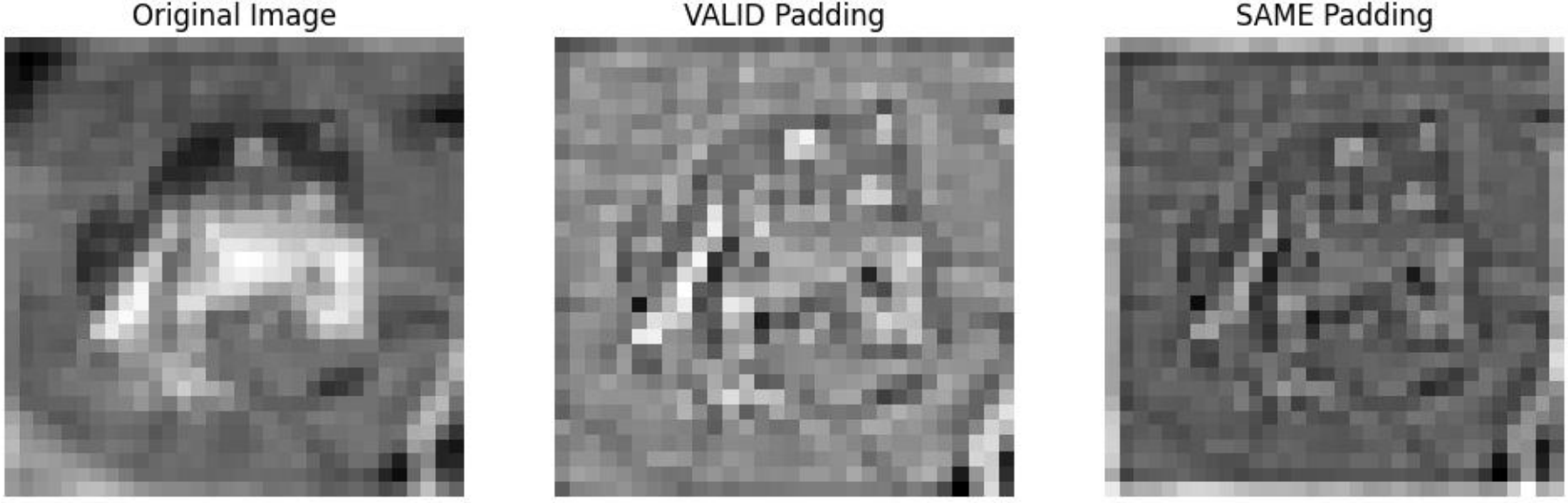


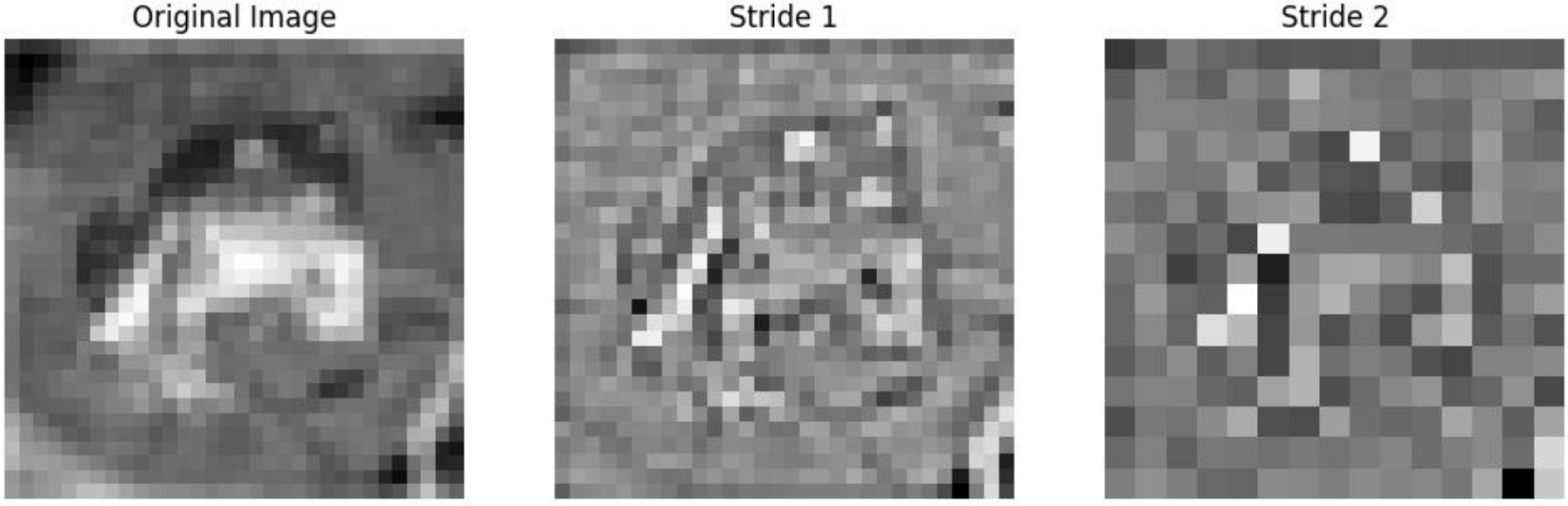

Original RGB Image

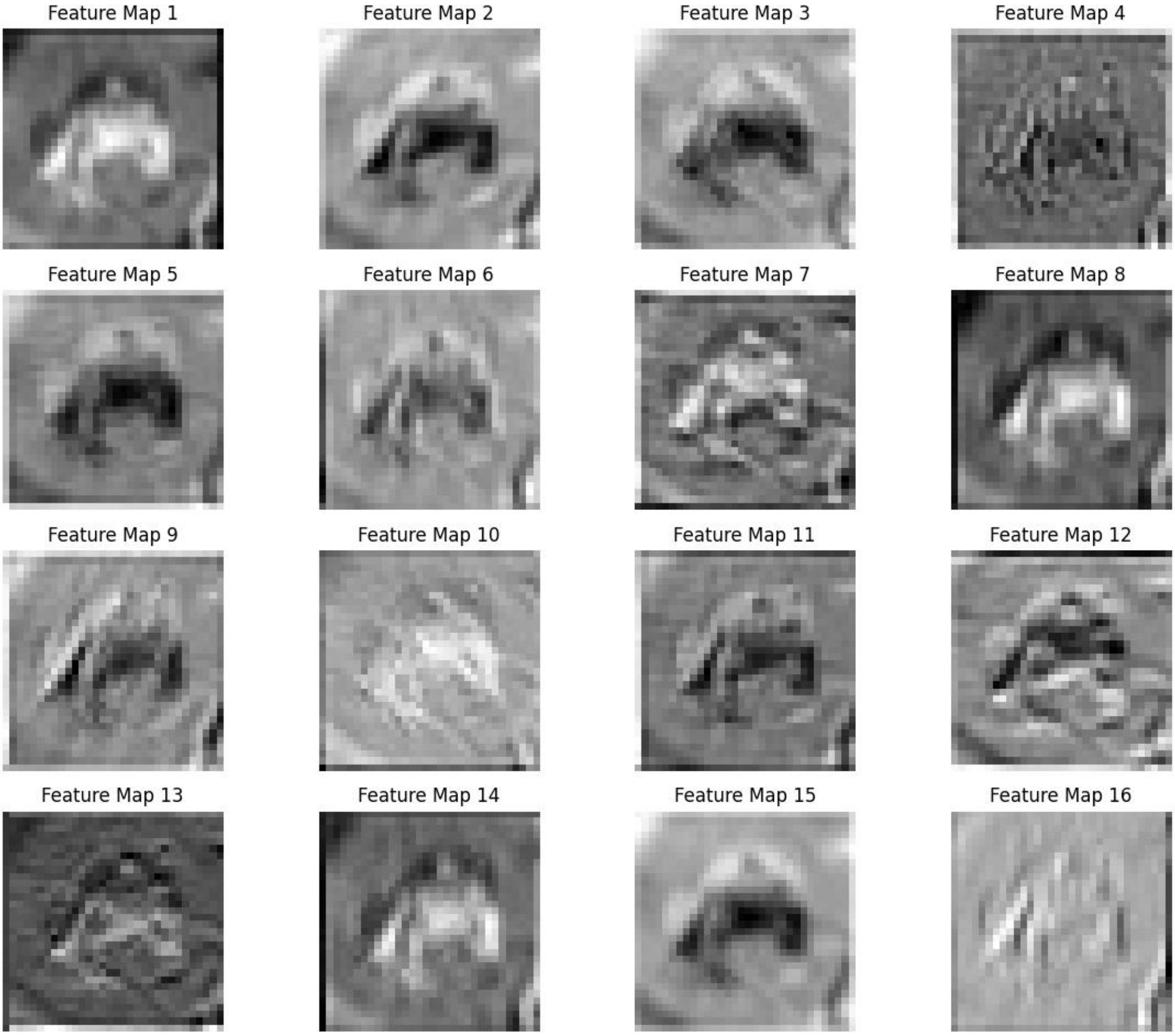

Grayscale Image


Original Grayscale Image

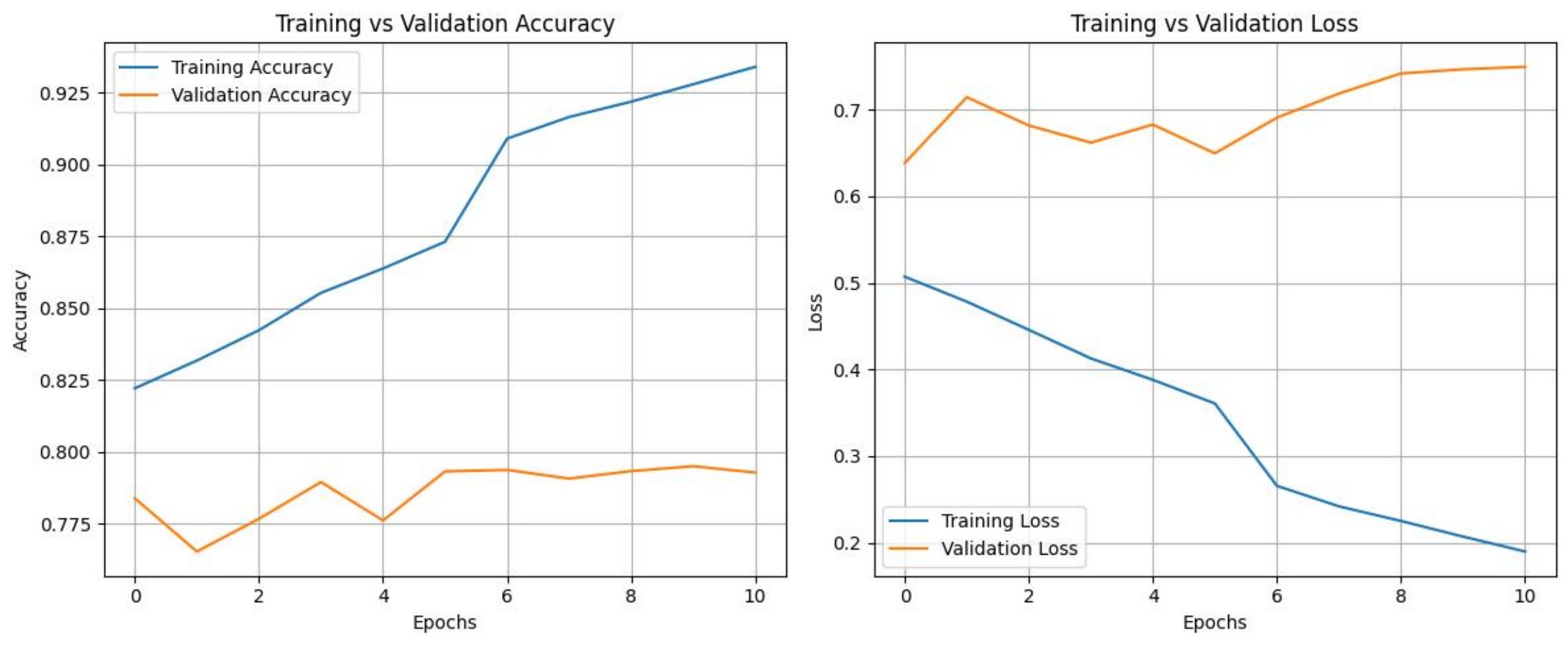

Edge Detection Image

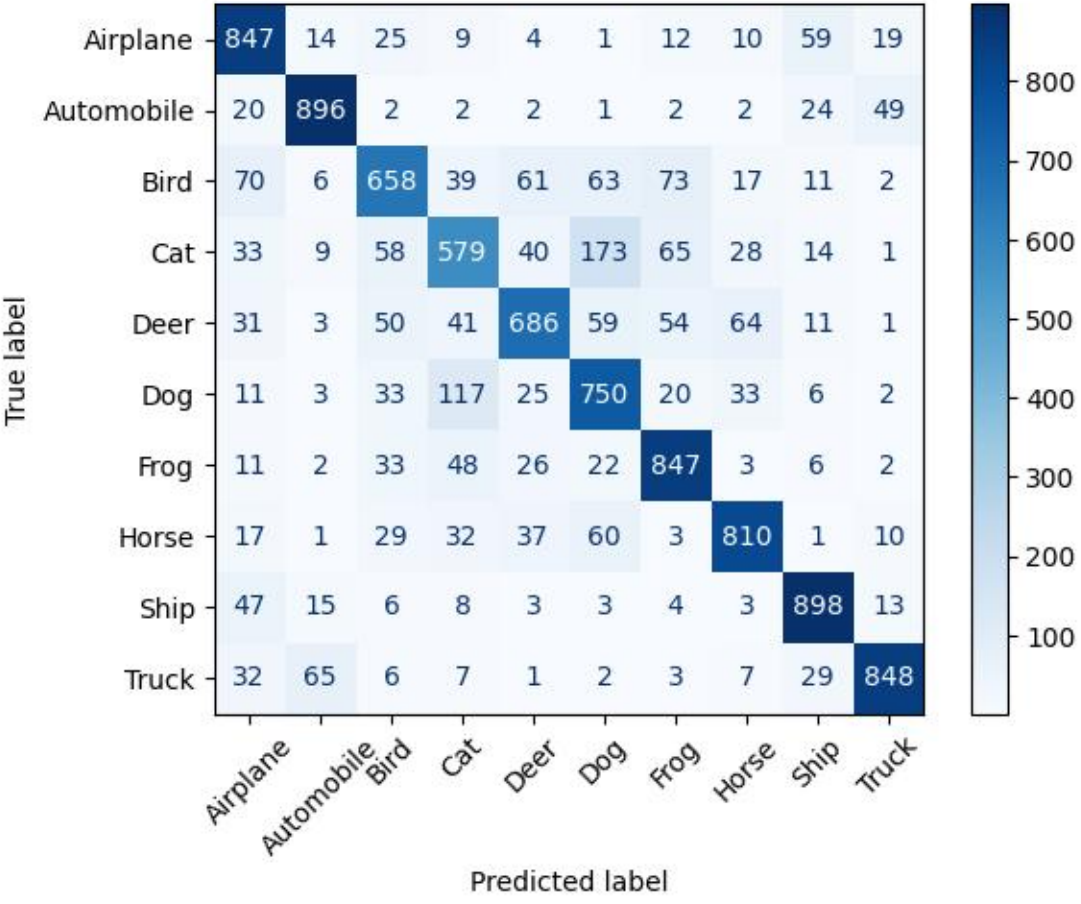

Original Image
VALID Padding
SAME Padding


Original Image
Stride 1
Stride 2


16 Convolution Feature Maps
Feature Map 1
Feature Map 2
Feature Map 3
Feature Map 4
Feature Map 5
Feature Map 6
Feature Map 7
Feature Map 8
Feature Map 9
Feature Map 10
Feature Map 11
Feature Map 12
Feature Map 13
Feature Map 14
Feature Map 15
Feature Map 16


Original Image
Max Pooling
Average Pooling


Training vs Validation Accuracy
Training Accuracy
Validation Accuracy
Accuracy
Epochs
0.925
0.900
0.875
0.850
0.825
0.800
0.775
0
2
4
6
8
10
Training vs Validation Loss
Training Loss
Validation Loss
Loss
Epochs
0.7
0.6
0.5
0.4
0.3
0.2


CIFAR-10 Confusion Matrix
Airplane 847 14 25 9 4 1 12 10 59 19
Automobile 20 896 2 2 2 1 2 2 24 49
Bird 70 6 658 39 61 63 73 17 11 2
Cat 33 9 58 579 40 173 65 28 14 1
Deer 31 3 50 41 686 59 54 64 11 1
Dog 11 3 33 117 25 750 20 33 6 2
Frog 11 2 33 48 26 22 847 3 6 2
Horse 17 1 29 32 37 60 3 810 1 10
Ship 47 15 6 8 3 3 4 3 898 13
Truck 32 65 6 7 1 2 3 7 29 848
Airplane
Automobile
Bird
Cat
Deer
Dog
Frog
Horse
Ship
Truck
True label
Predicted label
800
700
600
500
400
300
200
100

10 Misclassified CIFAR-10 Images



Original and Augmented CIFAR-10 Images

Original Image



Augmented Image 1



Augmented Image 2



Augmented Image 3



Augmented Image 4



Augmented Image 5



Augmented Image 6



Augmented Image 7

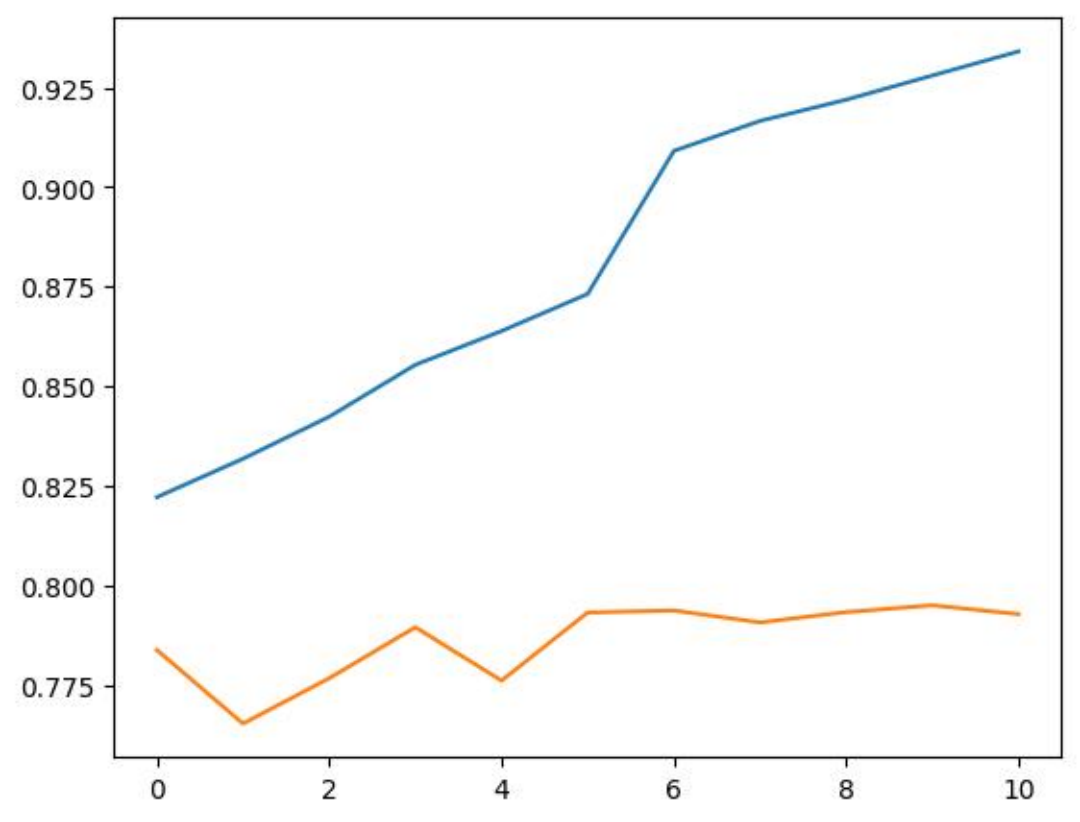


Augmented Image 8

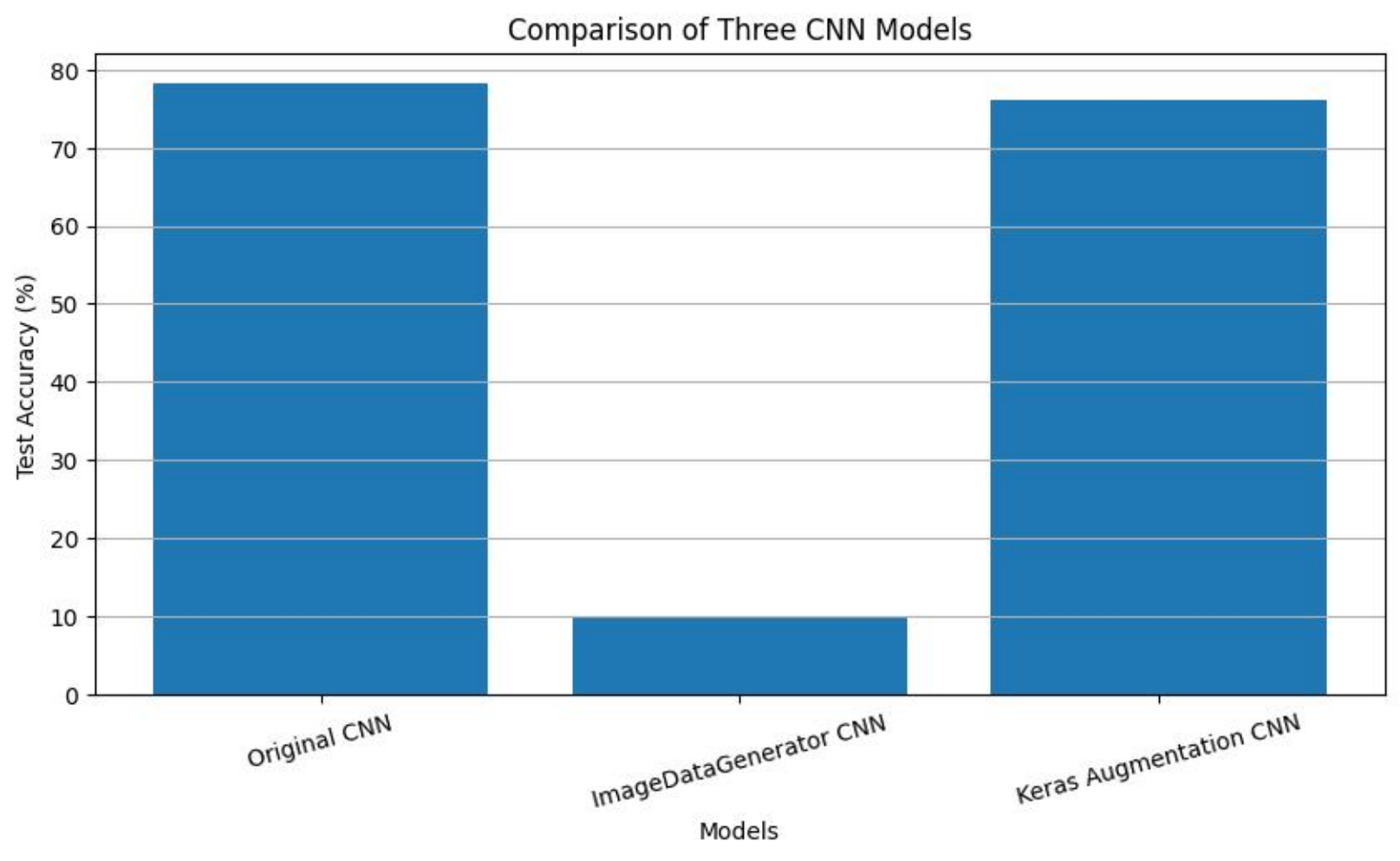


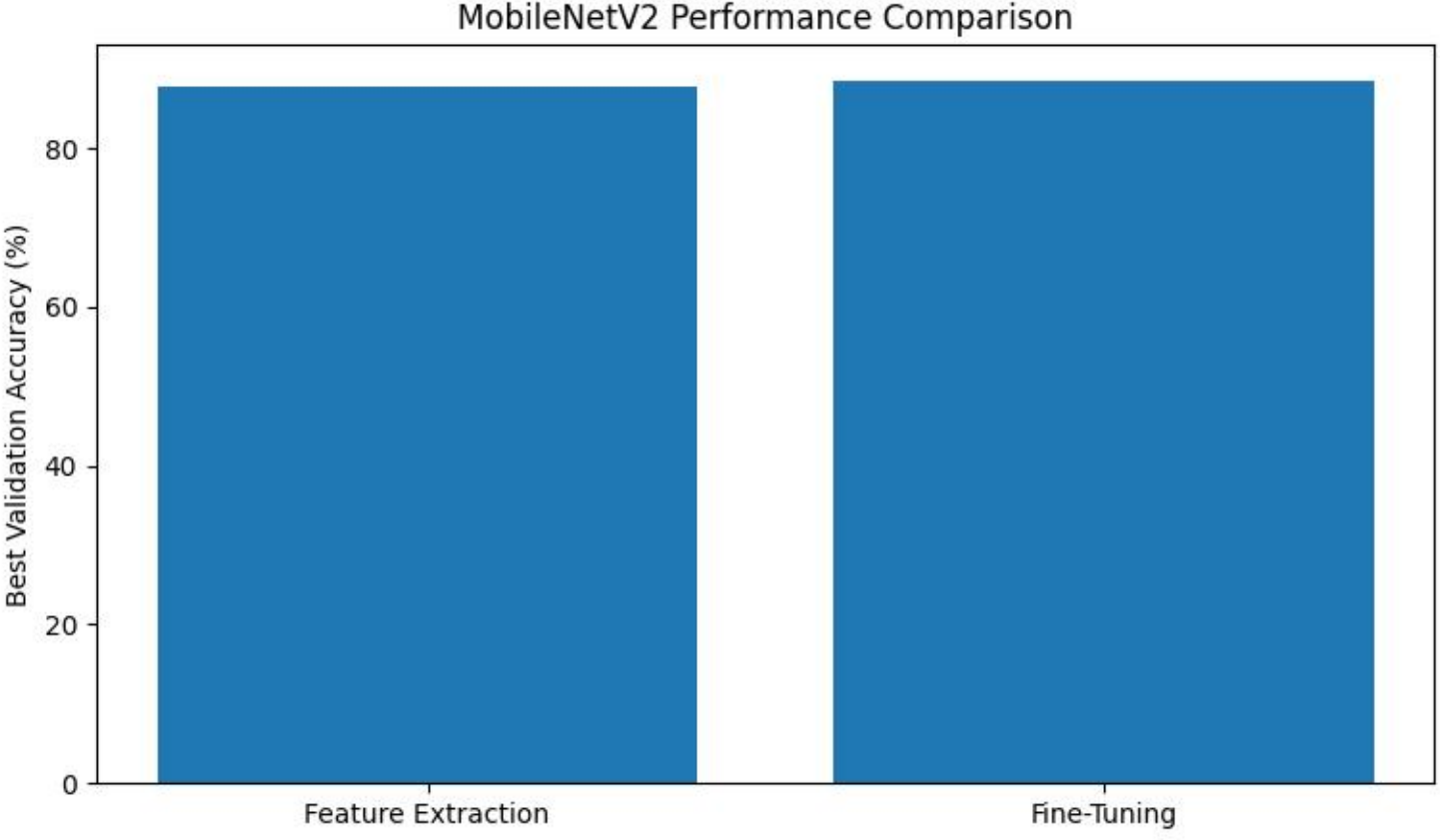
Augmented Image 9

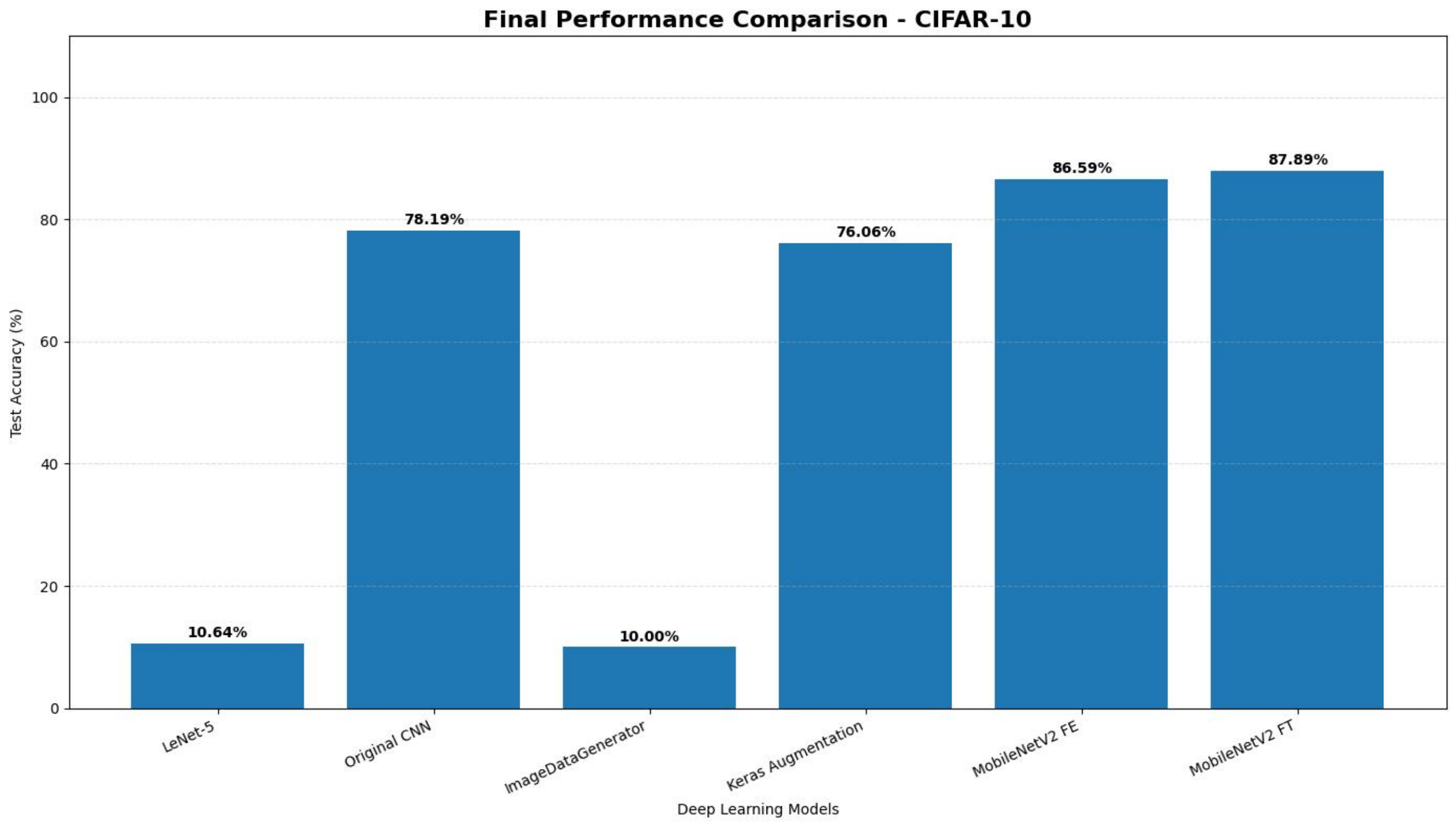



0.925
0.900
0.875
0.850
0.825
0.800
0.775
0
2
4
6
8
10


Keras Data Augmentation Examples
Augmented Image 1
Augmented Image 2
Augmented Image 3
Augmented Image 4
Augmented Image 5
Augmented Image 6
Augmented Image 7
Augmented Image 8
Augmented Image 9


Comparison of Three CNN Models
Test Accuracy (%)
0
10
20
30
40
50
60
70
80
Original CNN
ImageDataGenerator CNN
Keras Augmentation CNN
Models


MobileNetV2 Performance Comparison
Best Validation Accuracy (%)
0
20
40
60
80
Feature Extraction
Fine-Tuning


Final Performance Comparison - CIFAR-10
Test Accuracy (%)
0
20
40
60
80
100
10.64%
78.19%
10.00%
76.06%
86.59%
87.89%
LeNet-5
Original CNN
ImageDataGenerator
Keras Augmentation
MobileNetV2 FE
MobileNetV2 FT
Deep Learning Models